Compositional system

geometry and dynamics for,

S&P 500 GICS sector portfolio

day_000 → day_251

Report title	Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day
Generated	2026-05-06
Run ID	v1.1.x-final
Dataset ID	financial_sector
T (records)	252
D (carriers)	10
Engine version	cnt 2.0.4
Schema version	2.1.0
Engine signature (sha256[:32])	97748868a0381cb7b995d2ae562e5c42…
Source CSV SHA-256	b9fb10cd3fd290955e5b1adc14a9831f…
Closed data SHA-256	1520196588b2c76032ac3459e4c0ed2c…
Content SHA-256 (canonical JSON)	579724ce43356b2ab65a1d6294025719…

Reading order: geometry first, dynamics second.

```
§A  Data disclosure & processing notes
§B  Geometry : structure on the simplex
            evolution of proportions → ternaries → variation
            → correlation → scree → biplot → top-N biplot
            → balance-dendrogram → SBP table
§C  Dynamics : evolution through compositional space
            trajectories (Hs, ‖clr‖, ILR axes) → System Course Plot
            → bearing rose → bearing heatmap → ω → metric trace
            → helmsman → ring class → locks → CBS
            → pairwise divergence ranking → ω↔Hs
§D  Summary
```

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Plate 2 | §A Data Disclosure & Processing-Anomaly Review

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

```
INPUT DATA
  source file        : financial_sector_input.csv
  source SHA-256     : b9fb10cd3fd290955e5b1adc14a9831f64a45b264be85c31a3f9b765929c9dd
  closed data SHA-256: 1520196588b2c76032ac3459e4c0ed2cafe38e9a30690a729e37ec8c1b1f06a
  records (T)        : 252
  carriers (D)       : 10
  carriers           : Information Tech, Health Care, Financials, Cons Discretionary, Comm Services, Industrials, Cons Staples, Energy, Utilities, Materials

ORDERING
  is_temporal        : True
  ordering_method    : by-time

ZERO REPLACEMENT
  zeros replaced     : 0
  replacement δ      : 1e-15
  method             : multiplicative

UNITS LINEAGE  (schema 2.1.0)
  input_units                 : ratio
  higgins_scale_units         : neper
  units_scale_factor_to_neper : 1.0

ENGINE CONFIG ACTIVE OVERRIDES
  (none — defaults from cnt.py USER CONFIGURATION block)

ENGINE SOURCE FINGERPRINT
  stage2_locked.py      : c318ab82c15890b947c3fc15a6c66b55aa23f3cf94104cf9b9d5fbdb
  cnt.py                : 1e10cb62d6588ac9557c6a2e8436c2980f0c35b379f29c05c0377711
  sha256                : 97748868a0381cb7b995d2ae562e5c429988ee7fd556943b882c5fcb

DEGENERACY FLAGS
  (none)

LOCK EVENTS  (period-2 detector)
  total events       : 0

EITT BENCH-TEST RESIDUALS
  H_mean_full           : 2.0199146287814003
  M_sweep               : [{'M': 2, 'n_blocks': 126, 'H_mean_decimated': 2.0199167
  gate_pct              : 5.0
  note                  : Empirical observation of trajectory smoothness under tem

SANITY CHECKS  (computed during this report)
  ‖clr‖ non-finite count : 0
  ω    non-finite count : 0
  Hs   non-finite count : 0
  Σ CLR per row range   : [-6.22e-15, +6.66e-15]   (should be ≈ 0)
```

If any flag above is non-empty, the analysis on subsequent pages still ran but the reader should weight those plates accordingly. Determinism is preserved either way.

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Plate 3 | Evolution of Proportions (raw + cumulative)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

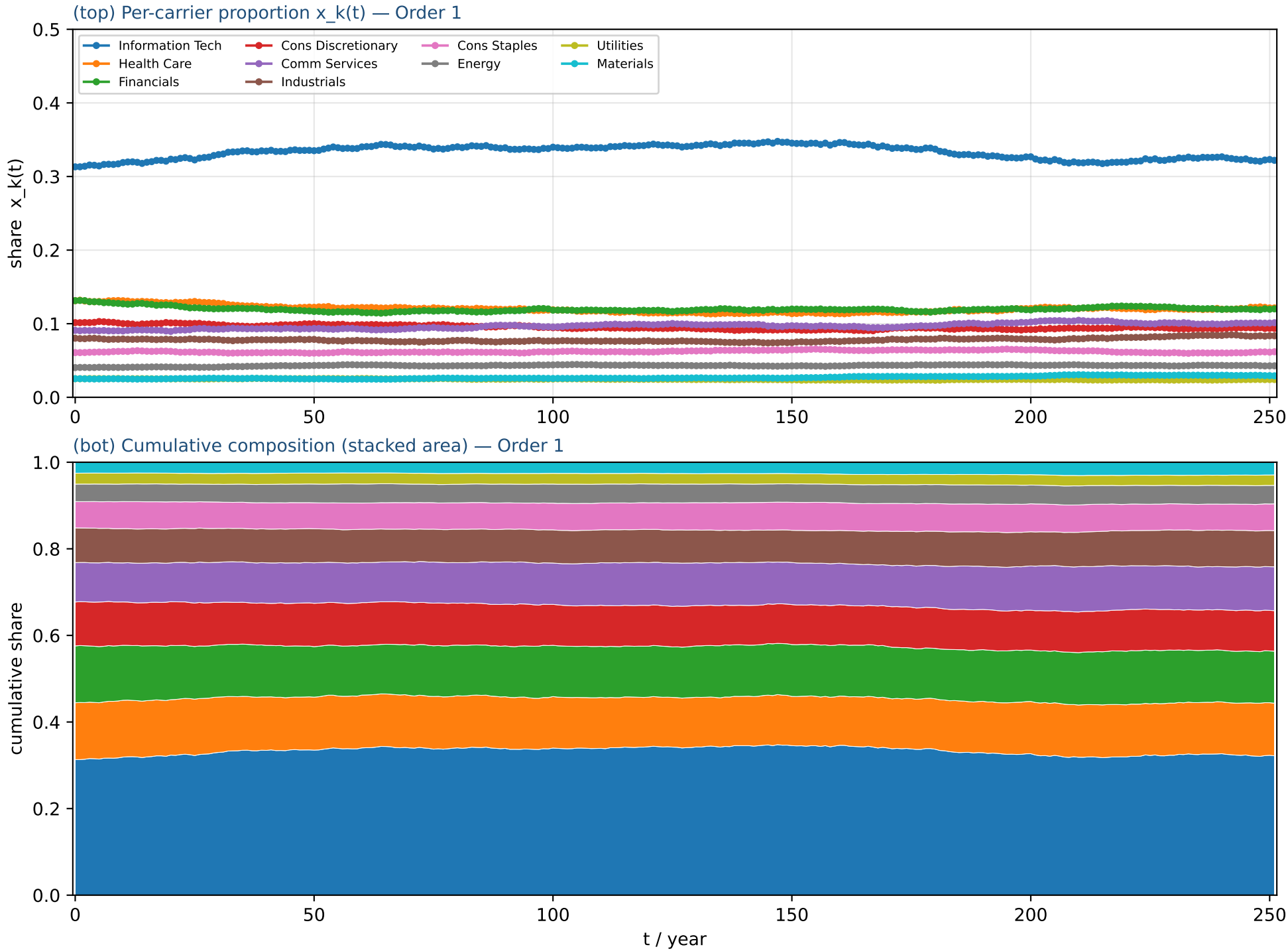

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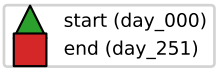
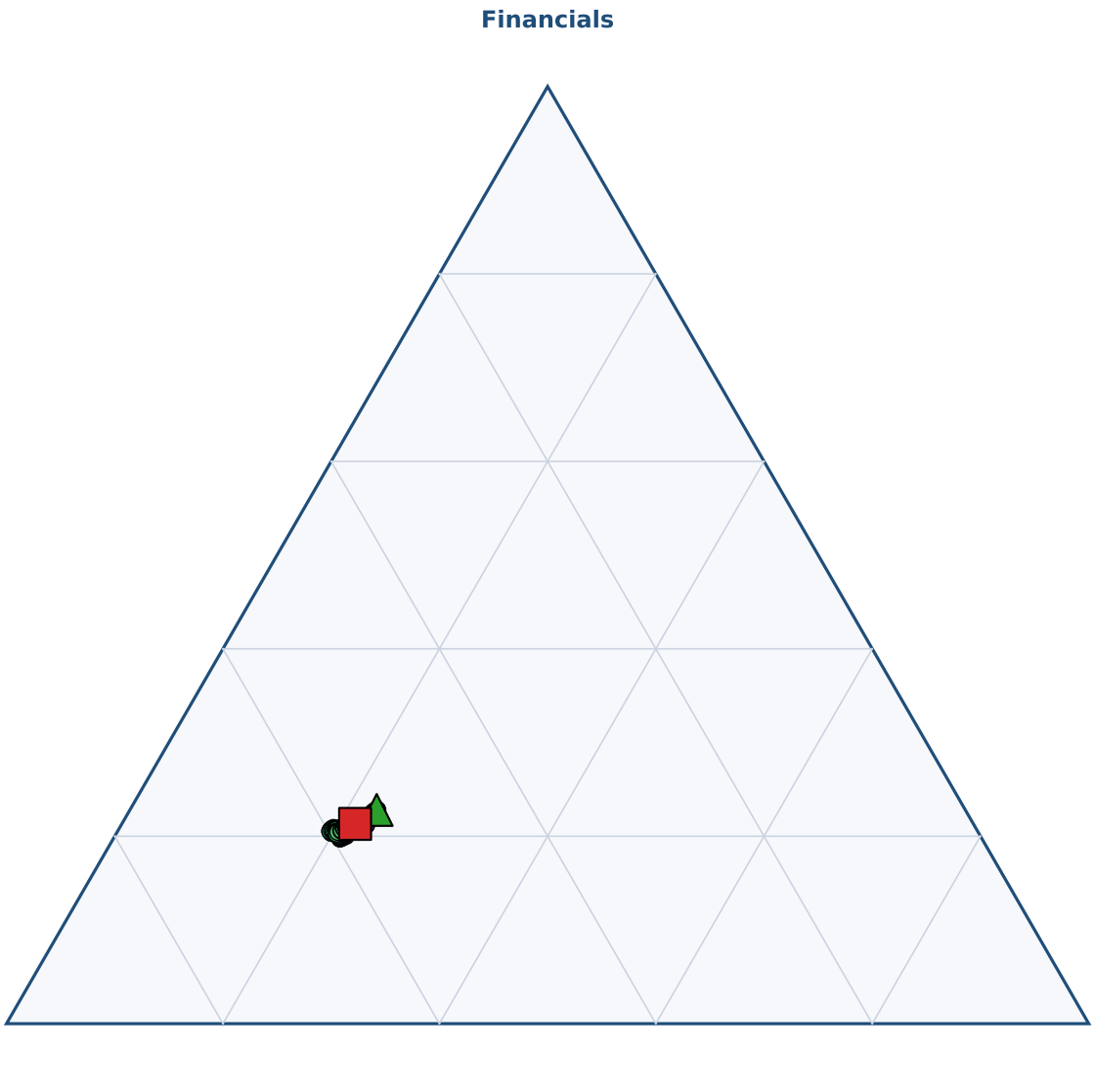
Plate 4 | Ternary Diagram (top-3 carriers)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

3-part subcomposition closed over Information Tech, Health Care, Financials
(remaining 7 carriers absorbed into closure)






```
Top-3 by mean share:
  Information Tech  33.3%
  Health Care     12.0%
  Financials      12.0%

T = 252 timesteps
colour map = time index
▲ start  ■ end
```

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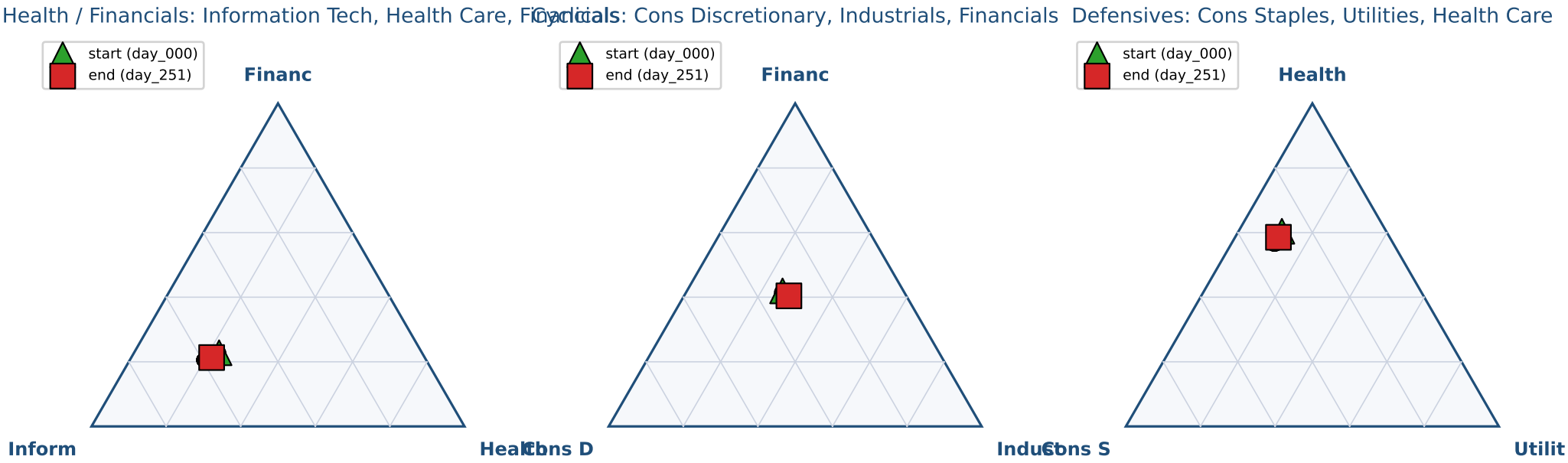
Plate 5 | Three-Part Subcompositions (3 ternaries)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Each ternary closes the chosen 3 carriers; remaining carriers absorbed into closure. ▲ start ■ end | colour map = time index

Triplet source: user-configured (master_control.json)

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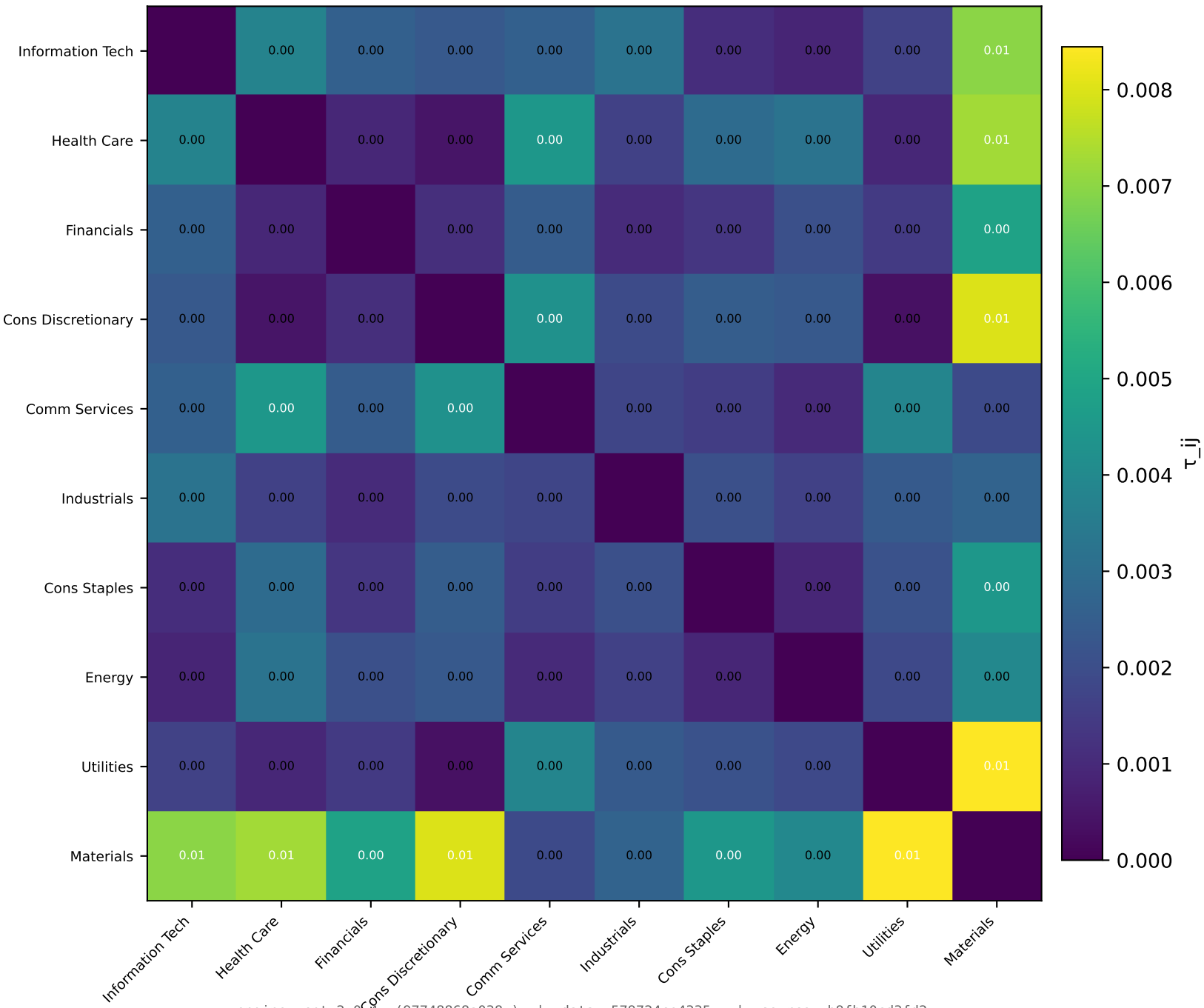
Plate 6 | Variation matrix τ_ij heatmap

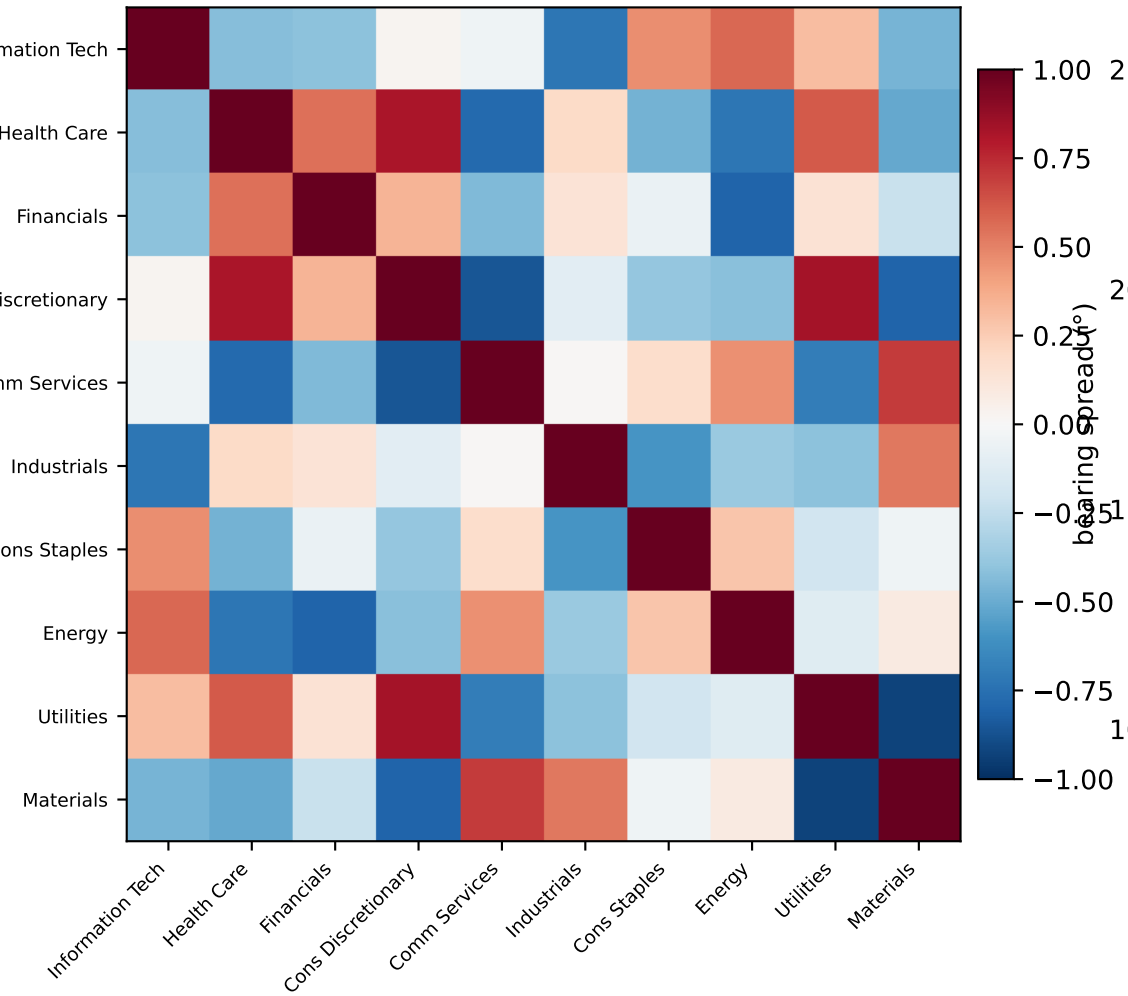
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

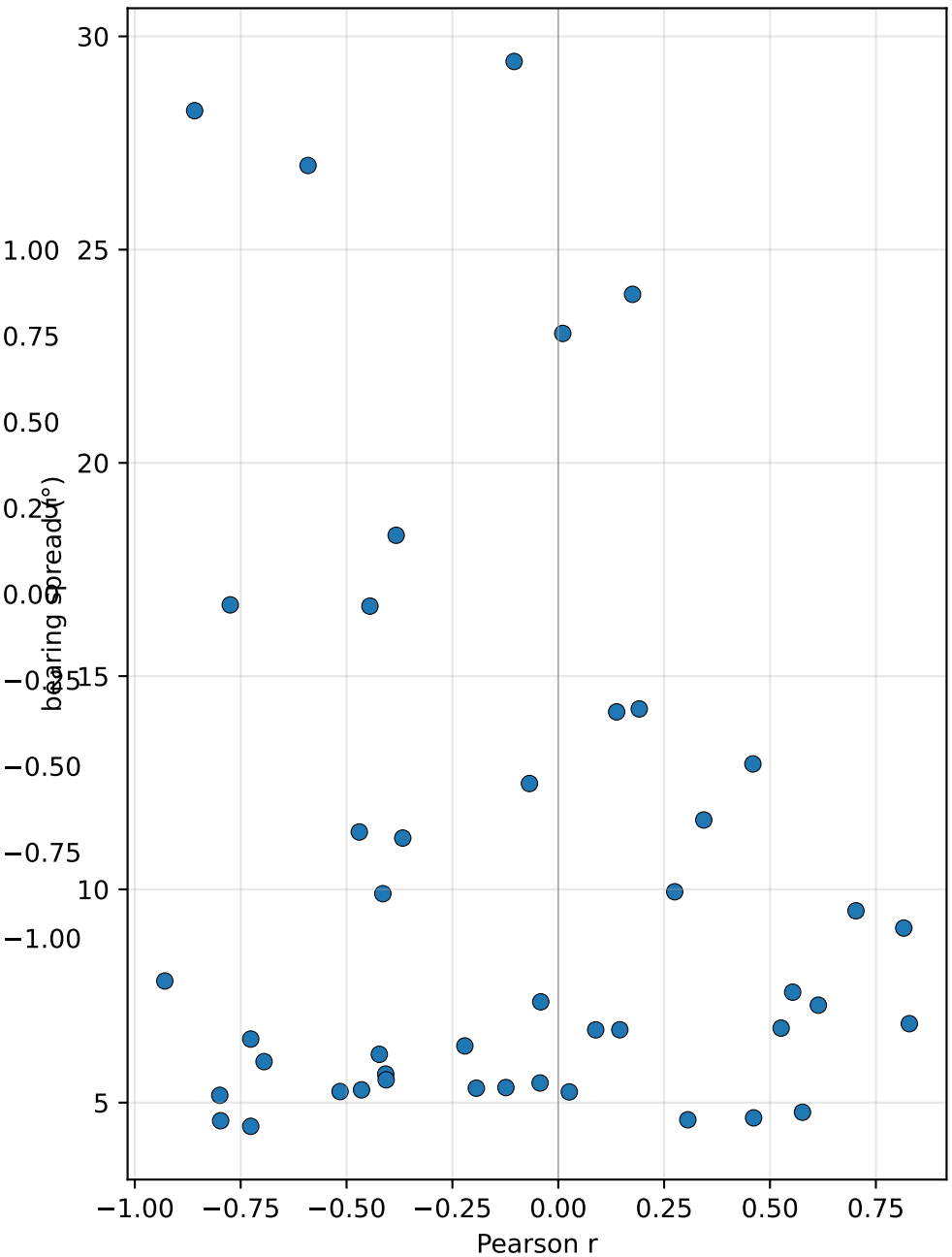


Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 7 | Pair Pearson r heatmap + co-movement scatter

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

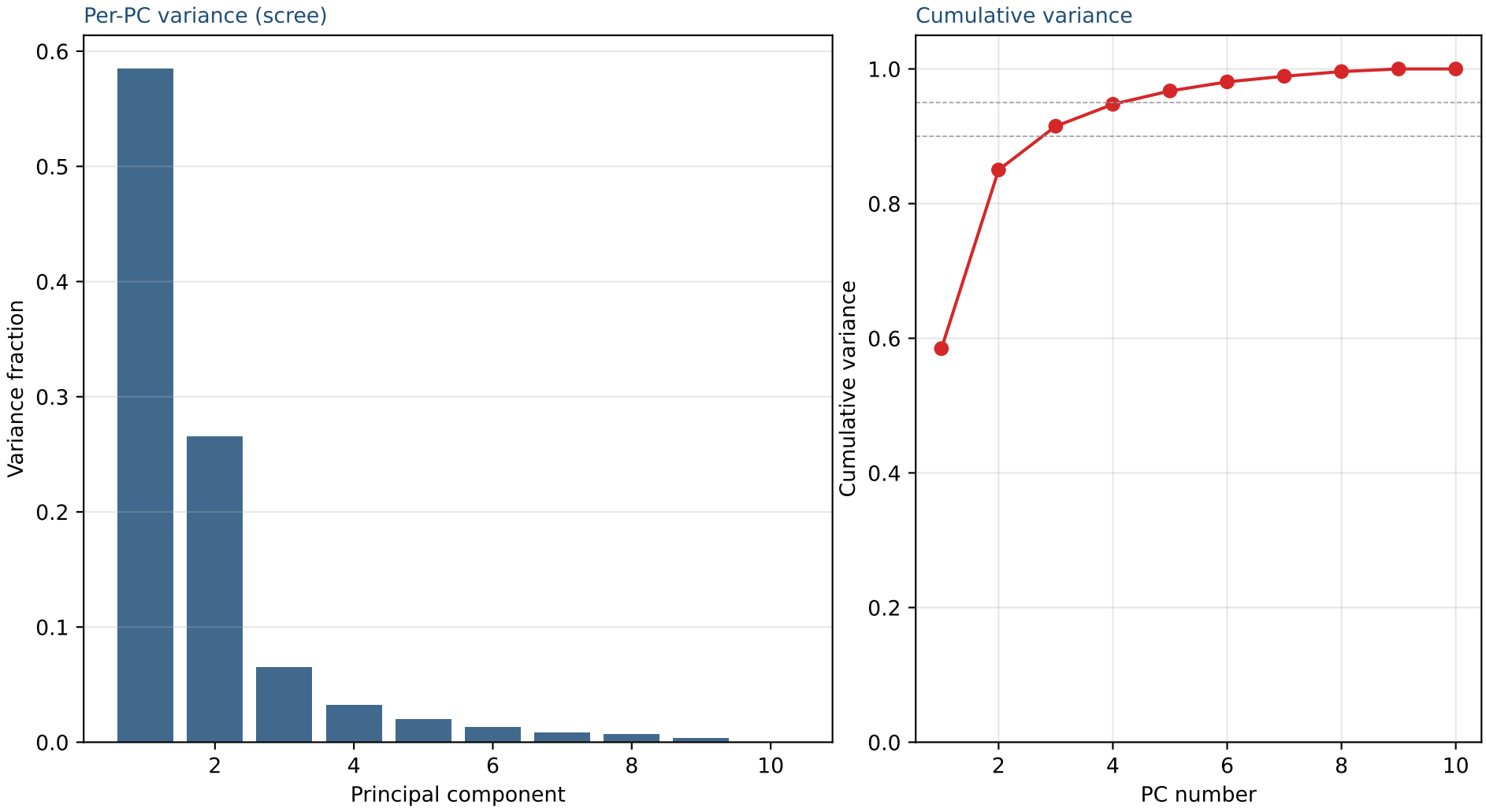




Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 8 | Scree Plot - CLR PCA

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2




```
Per-PC variance contribution:
  PC1: 58.46%   cum 58.46%
  PC2: 26.54%   cum 85.00%
  PC3:  6.50%   cum 91.49%
  PC4:  3.25%   cum 94.75%
  PC5:  1.99%   cum 96.73%
  PC6:  1.35%   cum 98.08%
  ... 4 more PCs
```

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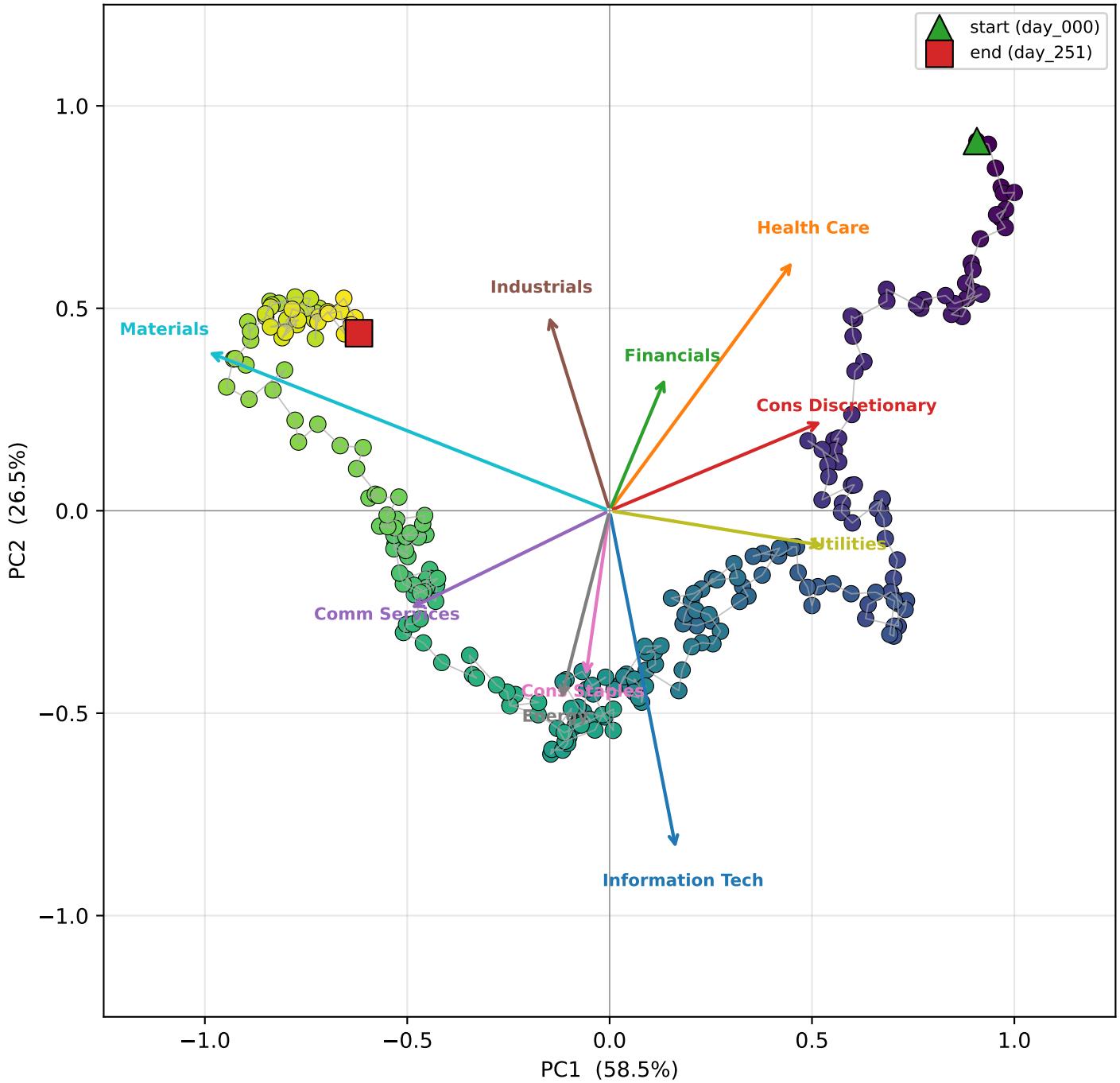
Plate 9 | CLR Biplot (CoDa biplot)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

CoDa biplot - covariance form, 2D variance explained 85.0%




```
CoDa biplot (Aitchison-Greenacre):
  - Points: timestep scores in PC(1,2) of c
  - Rays:   carrier loadings (columns of V)
  - Ray length = carrier importance on the
  - Angle between rays ~ negative log-ratio
  - Distance between point projections = ap
    of the Aitchison distance restricted to

PC1 captures 58.5% of variance.
PC2 captures 26.5% of variance.
Total 2D plane:  85.0%
```

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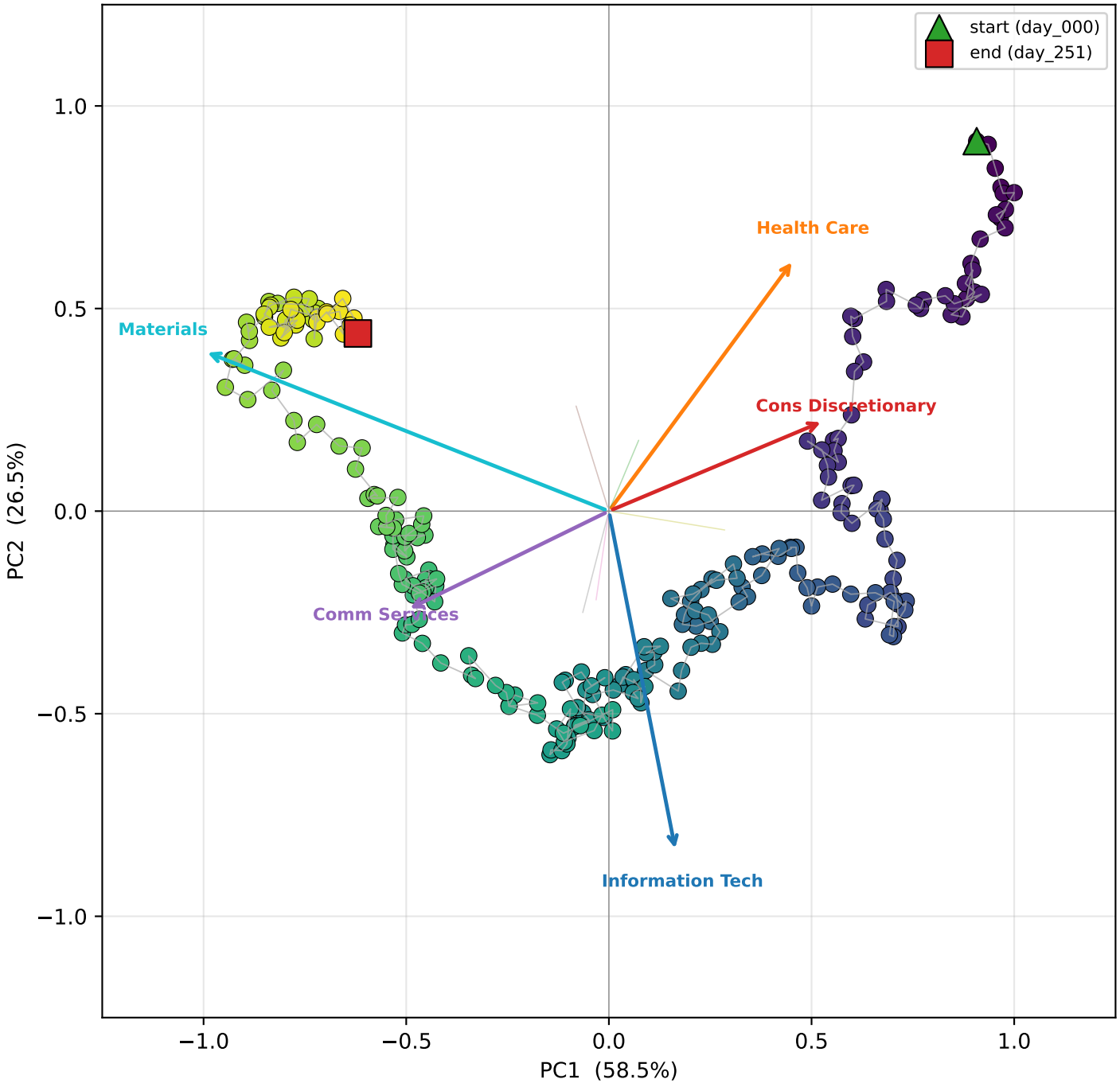
Plate 10 | CLR Biplot — top-5 carriers by loading magnitude

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Top-5 biplot — 2D variance 85.0% (D = 10, dropped 5 short rays)



Loading magnitudes $\sqrt{(G_1^2 + G_2^2)}$:

- Materials 0.7381
- Information Tech 0.5895
- Health Care 0.5304
- Cons Discretionary 0.3964
- Comm Services 0.3822

Dropped (faint stubs only):

Utilities	0.3745
Industrials	0.3521
Energy	0.3358
Cons Staples	0.2895
Financials	0.2498

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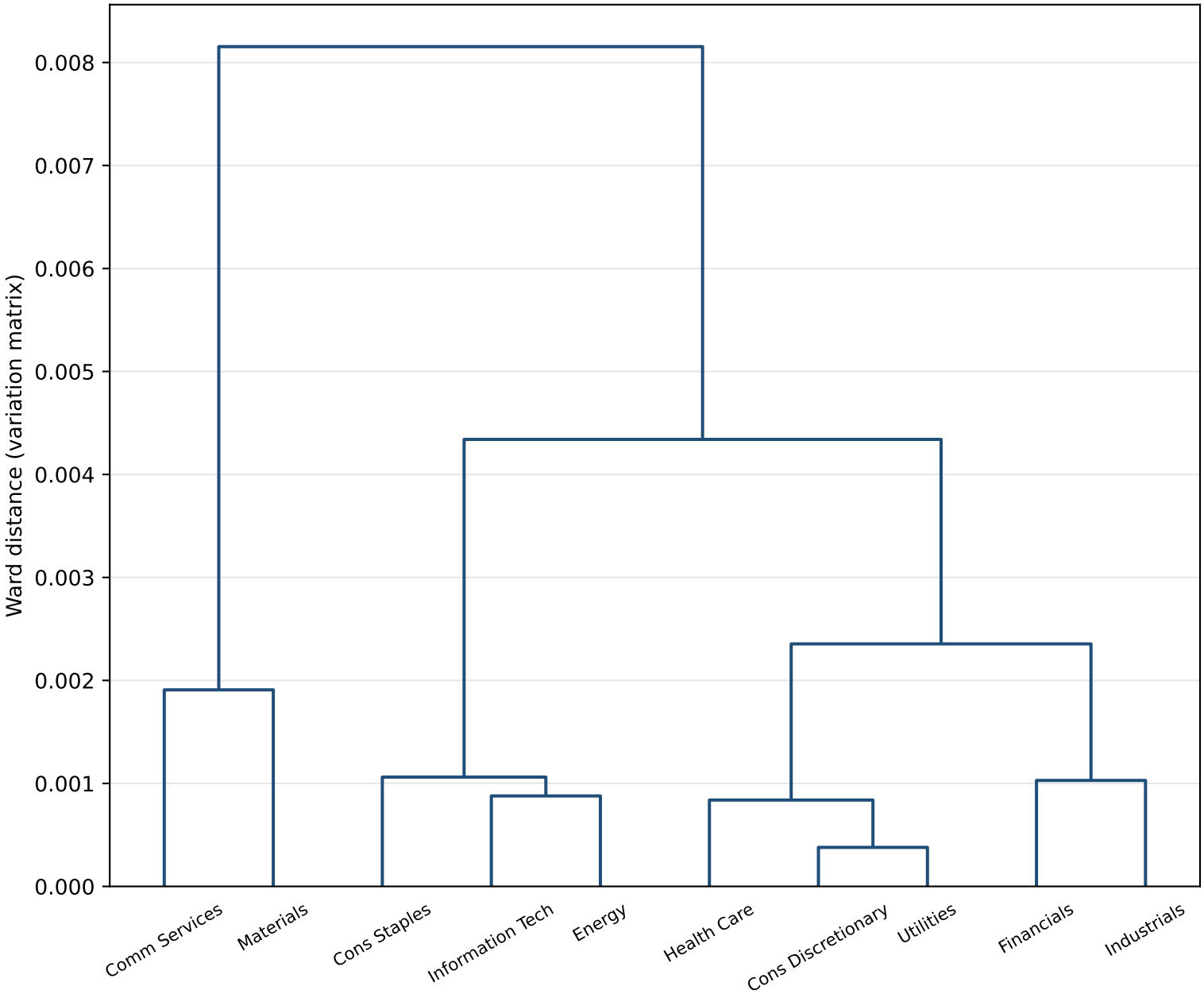
Plate 11 | Balance-Dendrogram (Ward on variation matrix)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Hierarchical balance tree — each merge defines one balance partition




```
Balance-dendrogram (Pawlowsky-Glahn-
  – Ward clustering on the (symmetri
  – Each internal node merges two ca
  – That merge defines a balance:
      b = sqrt(|A||B|/(|A|+|B|)) ·
    where g() is the geometric mean.
  – The balances form an orthonormal

D = 10 carriers → D-1 = 9 balances.
Tightly-clustered carriers (low vari
See companion plate for the full SBP
```

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Plate 12 | Sequential Binary Partition (SBP) Matrix

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

SBP cells: + = numerator, - = denominator, . = not in this balance

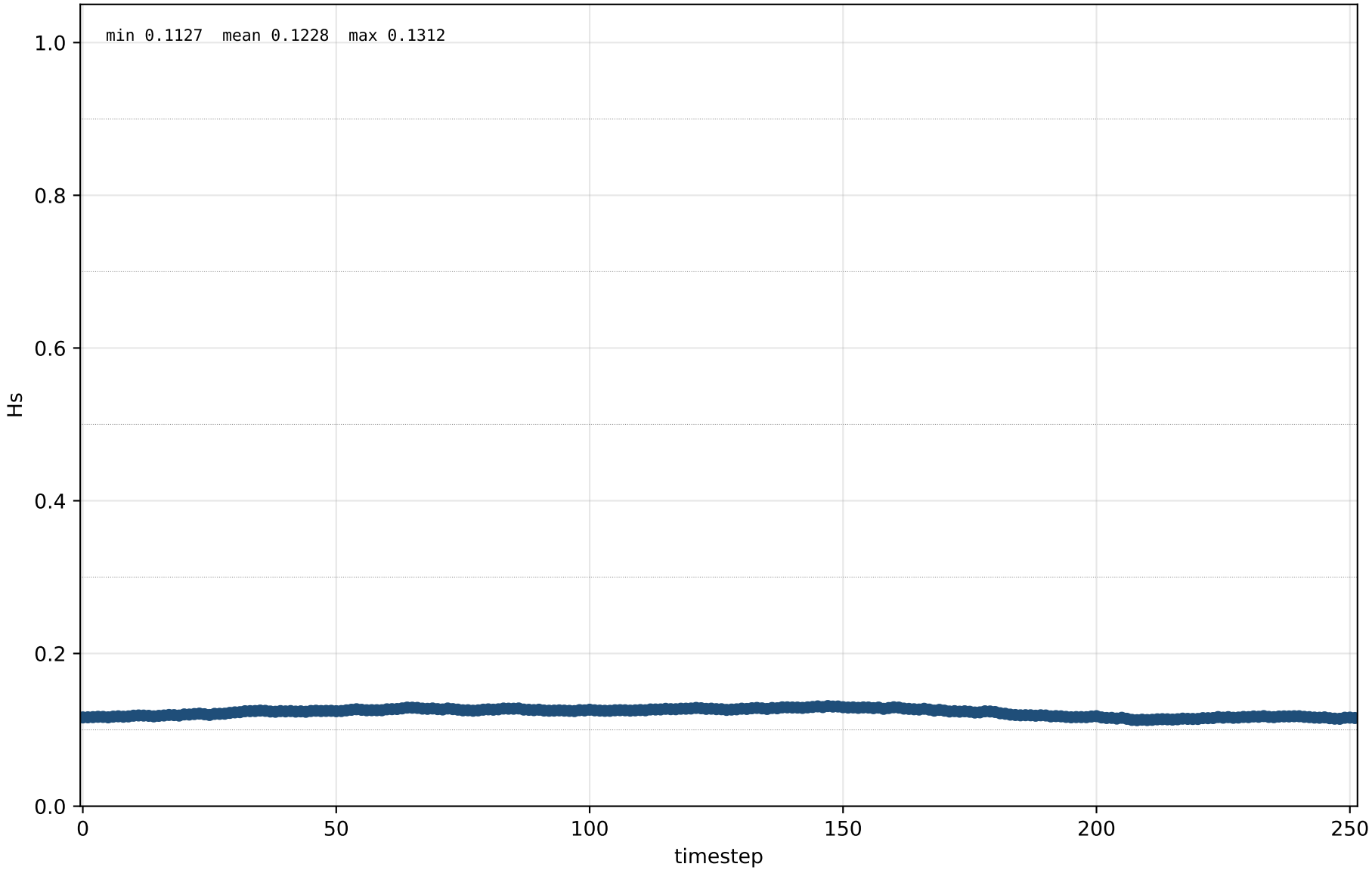
	b1	b2	b3	b4	b5	b6	b7	b8	b9
on Tech	.	.	+	.	-	.	.	+	-
th Care	.	+	.	.	.	.	+	-	-
ancials	.	.	.	+	.	.	-	-	-
tionary	+	-	.	.	.	.	+	-	-
ervices	.	.	.	.	.	+	.	.	+
strials	.	.	.	-	.	.	-	-	-
Staples	.	.	.	.	+	.	.	+	-
Energy	.	.	-	.	-	.	.	+	-
lities	-	-	.	.	.	.	+	-	-
terials	.	.	.	.	.	-	.	.	+

Balance b_k = sqrt(|A_k||B_k|/(|A_k|+|B_k|)) * ln(g(A_k) / g(B_k)) | D = 10; D-1 = 9 balances.

Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 13 | Hs (Higgins scale) trajectory — Order 2 aggregate

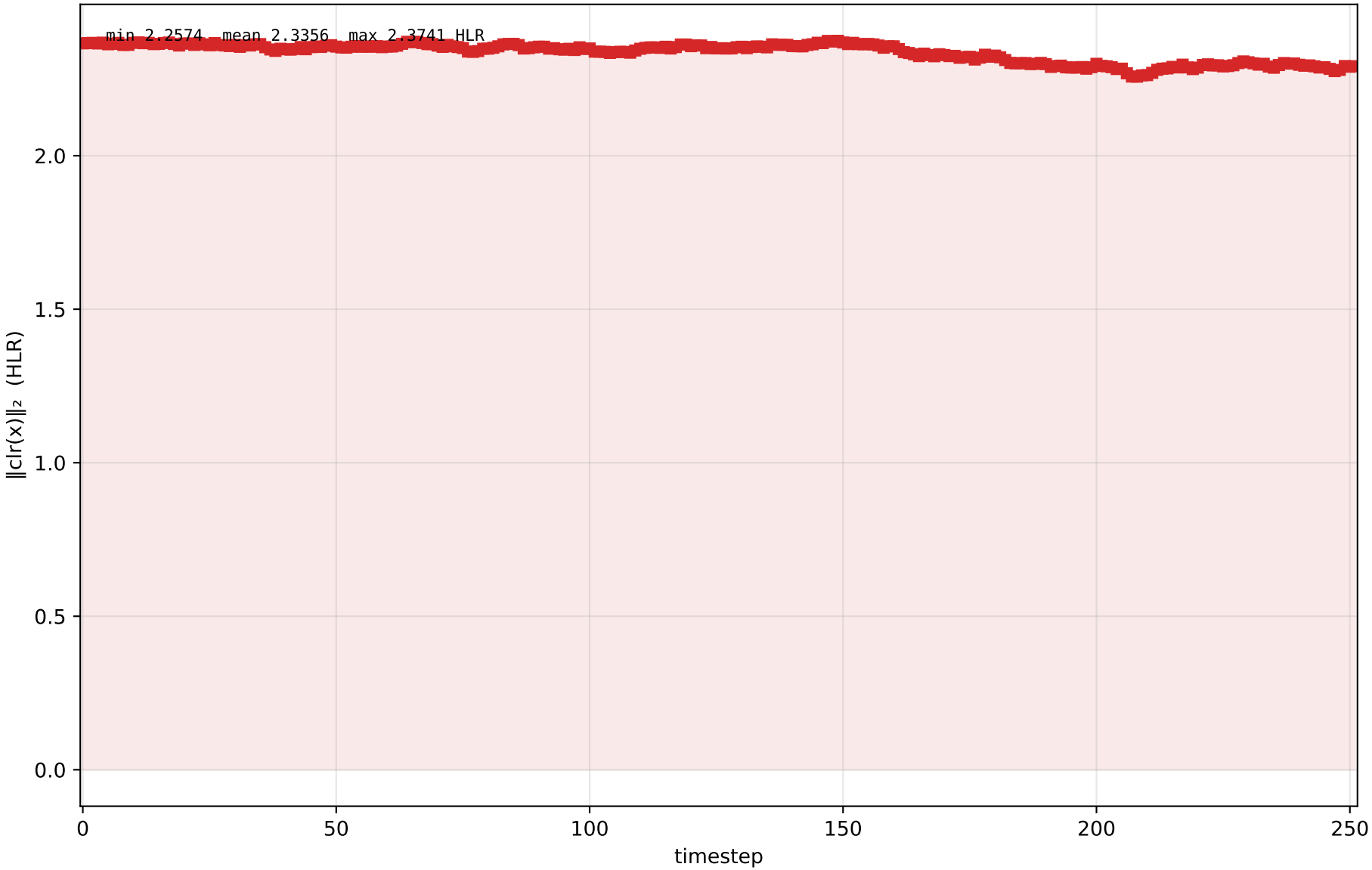
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 14 | Aitchison norm trajectory ‖clr(x)‖₂ — Order 2 aggregate

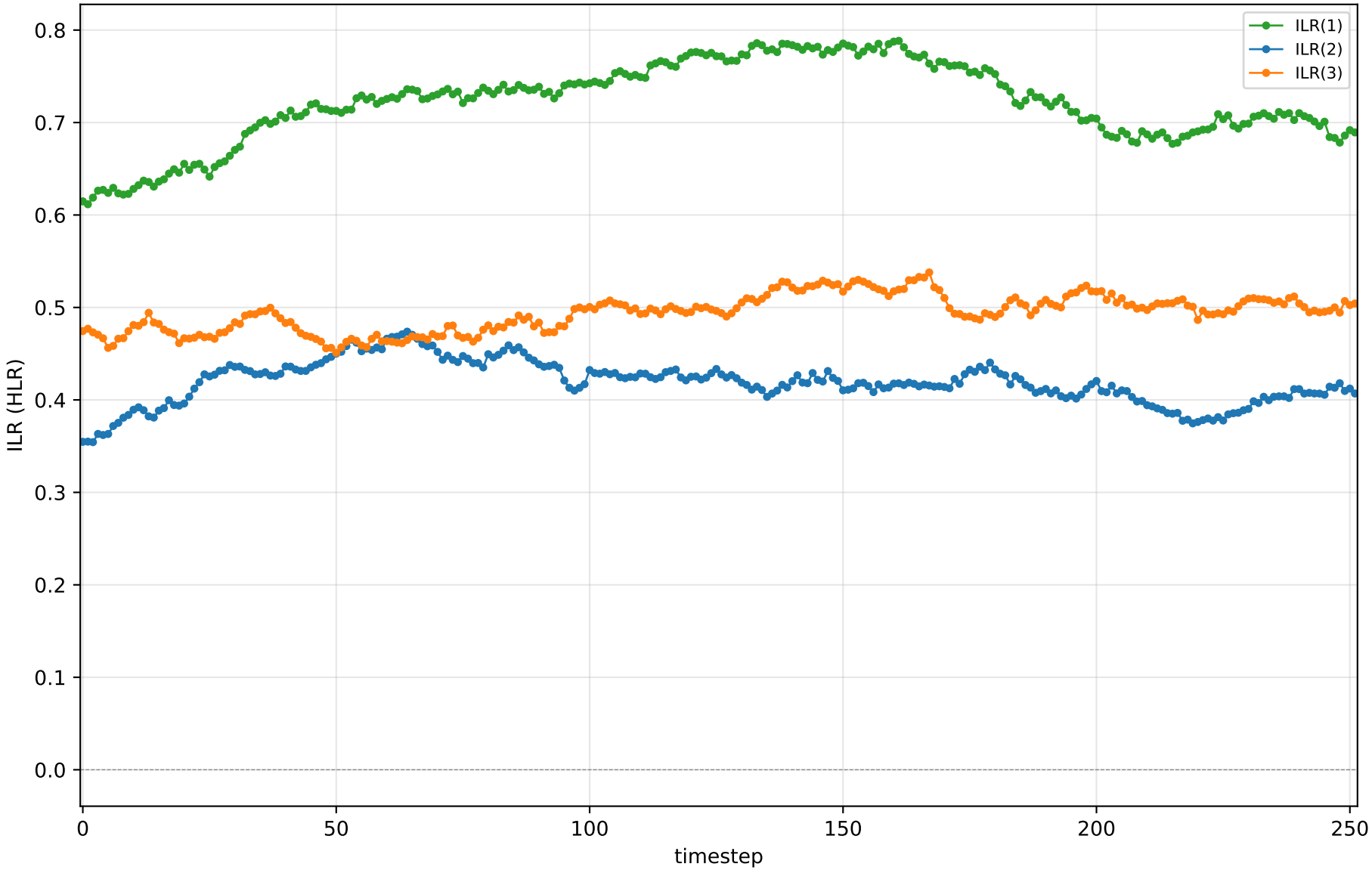
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 15 | ILR axes 1–3 trajectories (Helmert basis) — Order 2 aggregate

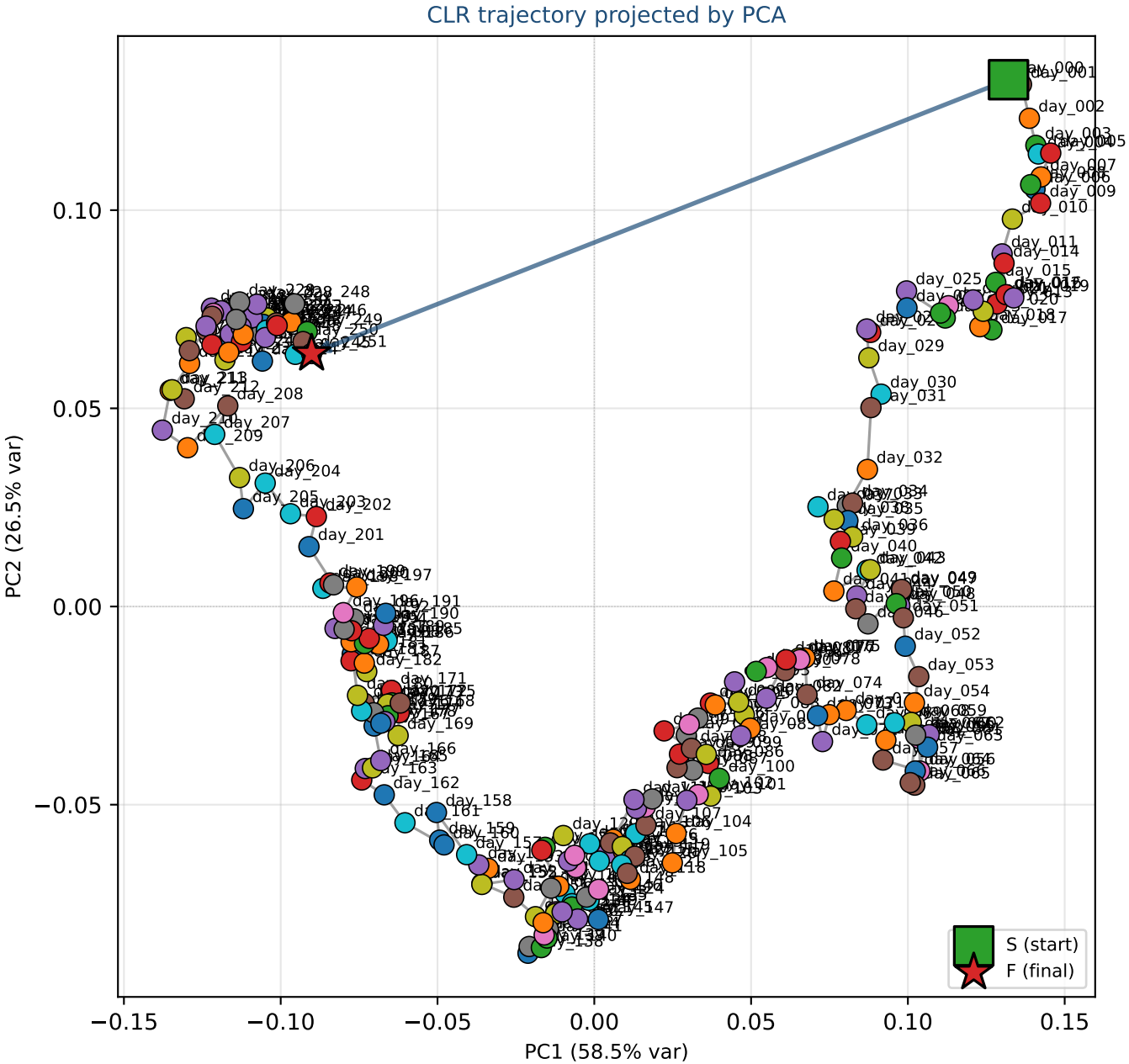
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



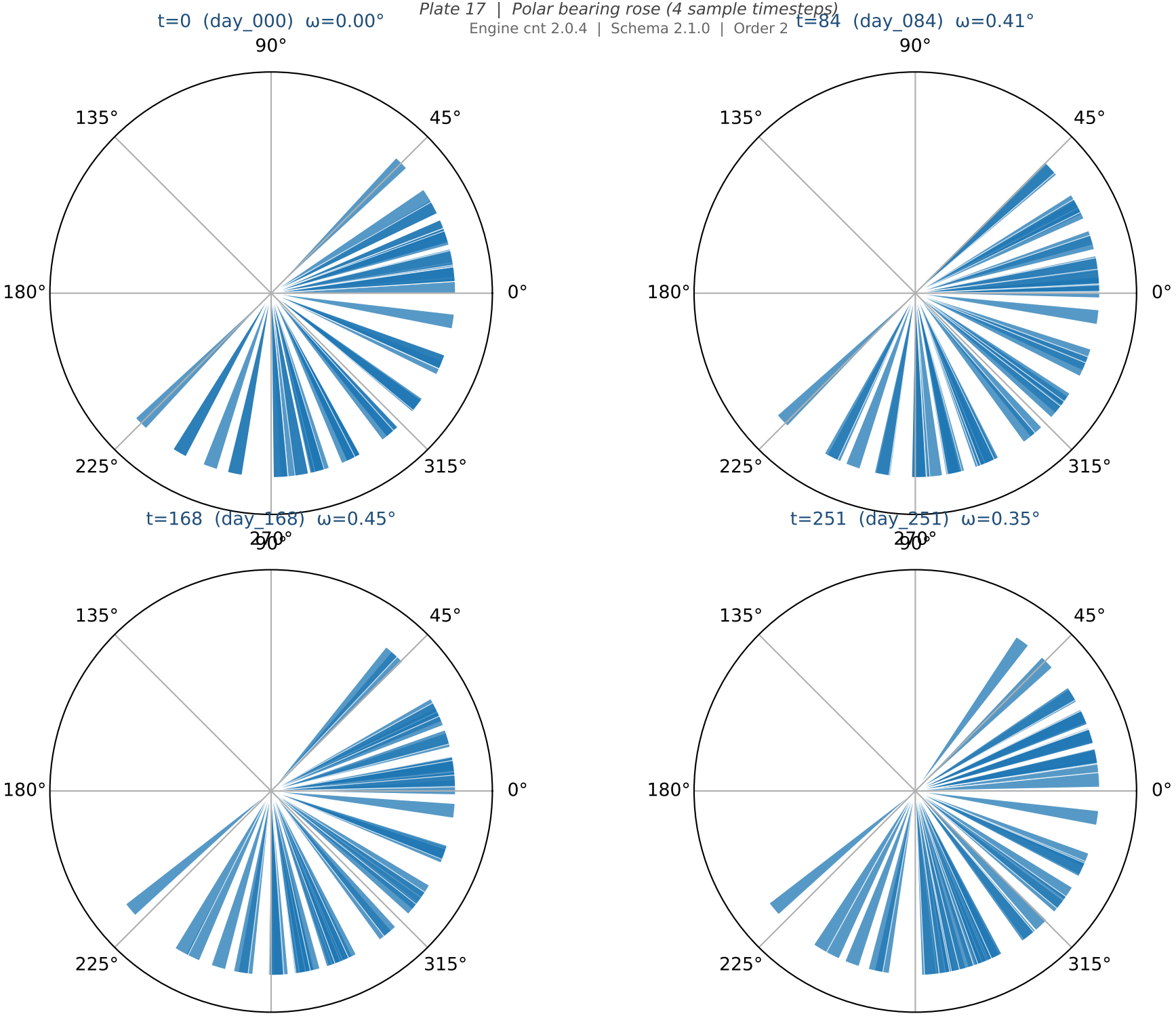
Navigation metrics (V_net = h_F − h_S)

```
h_start label    : day_000
h_final label    : day_251

net_distance     : 0.2452 HLR
path_length      : 3.6936 HLR
course_directness: 0.0664
  (1.0 = straight line, 0 = pure looping)

dynamic range S  : 2.5182 HLR
dynamic range F  : 2.5909 HLR

Reading guide:
  tight cluster  → stable composition
  long excursion → structural shift
  sharp turn     → regime transition
  loop           → cyclic behaviour
  spiral         → systematic drift
```

Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 17 | Polar bearing rose (4 sample timesteps)

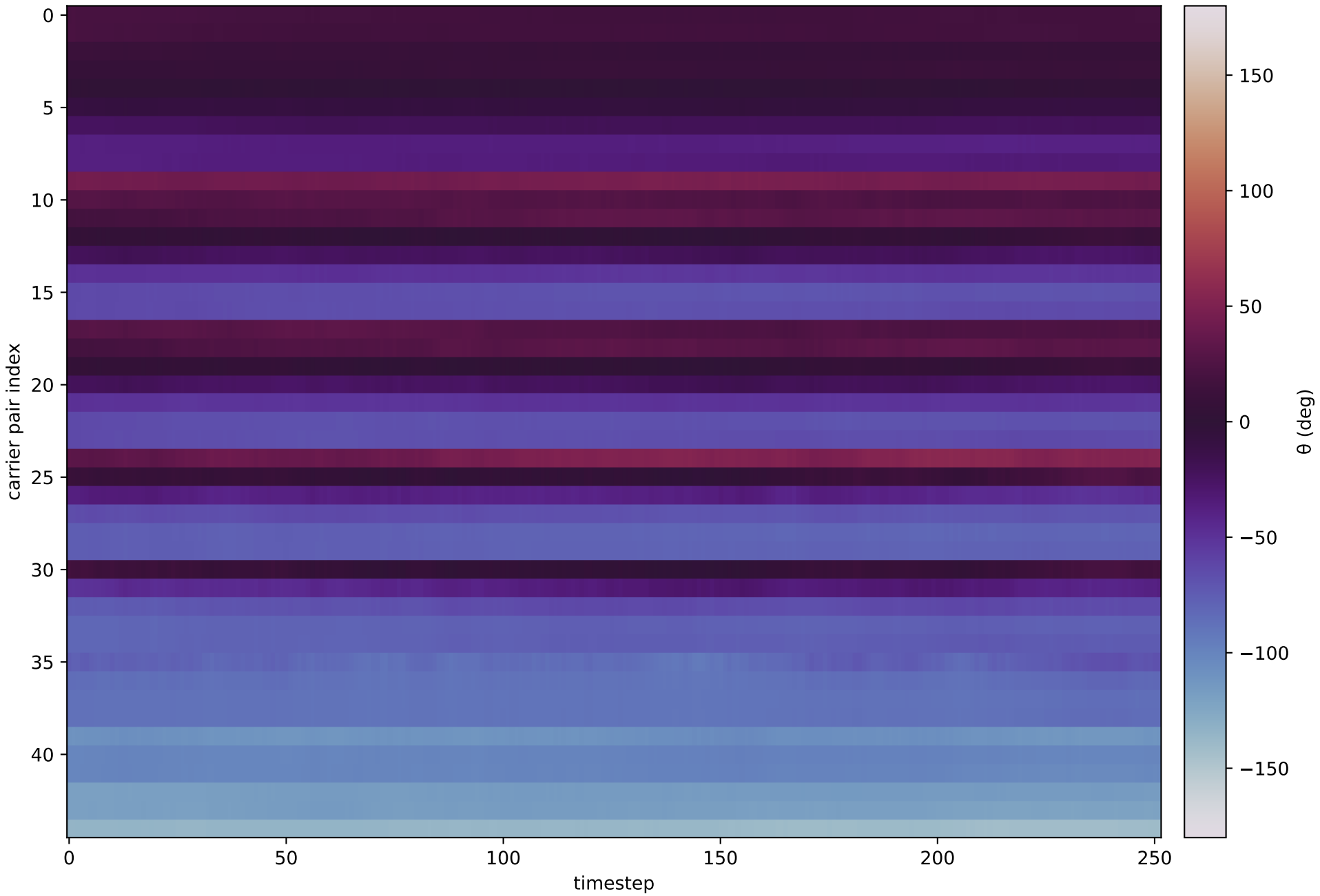
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 18 | Bearing heatmap T × C(D,2) pairs

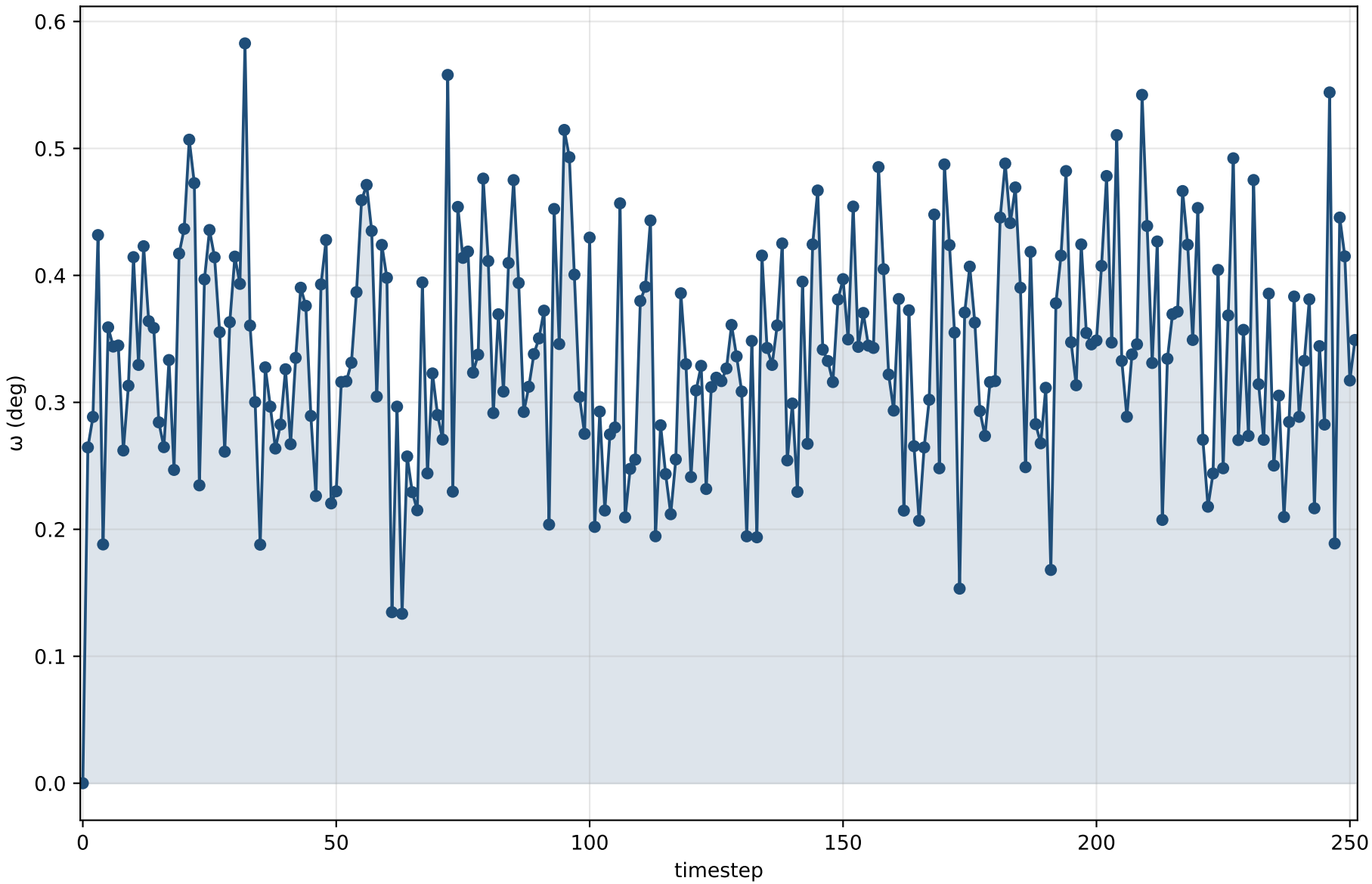
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 19 | ω angular velocity trajectory

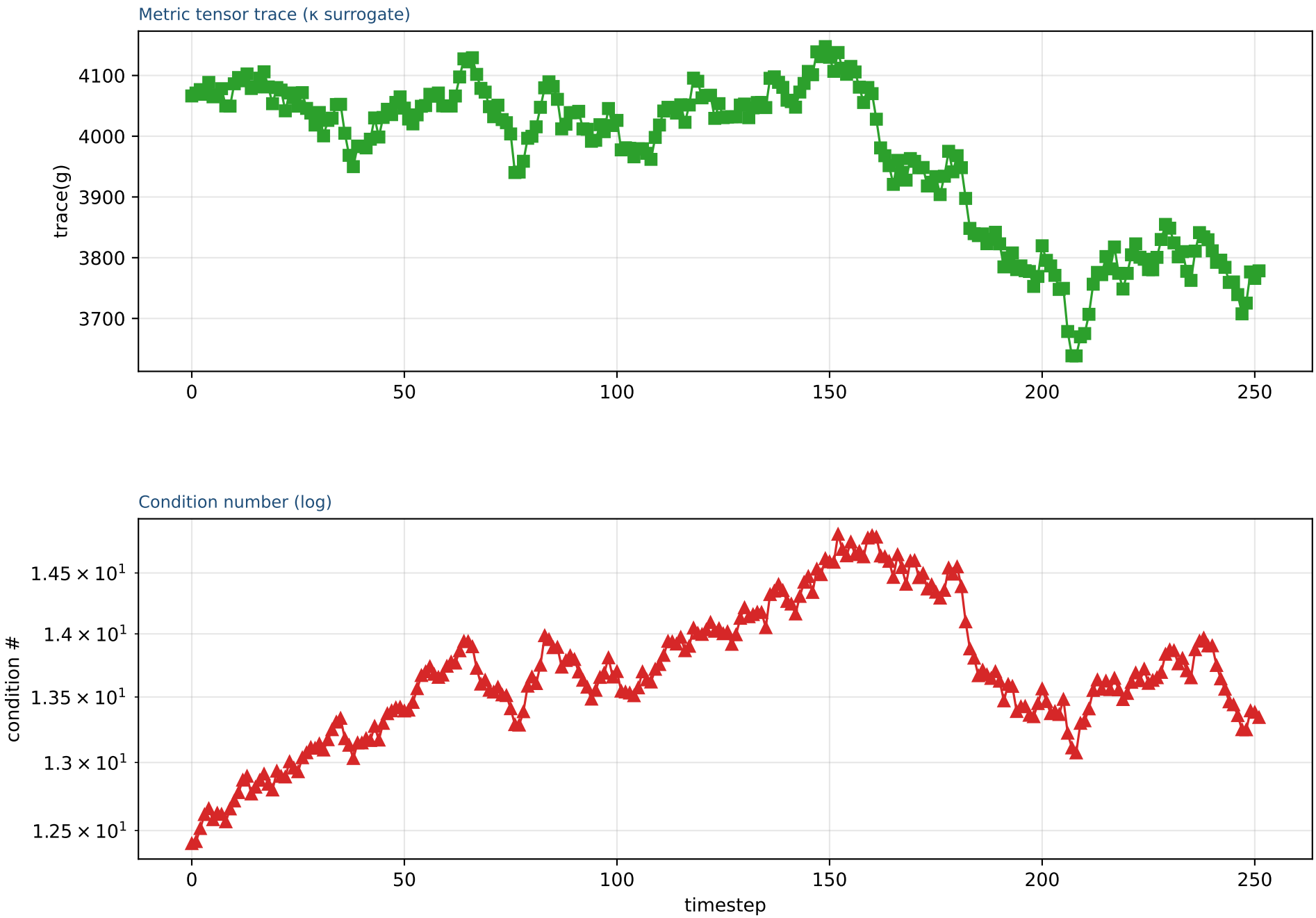
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 20 | Metric tensor trace + condition number

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 21 | Helmsman σ lineage timeline

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

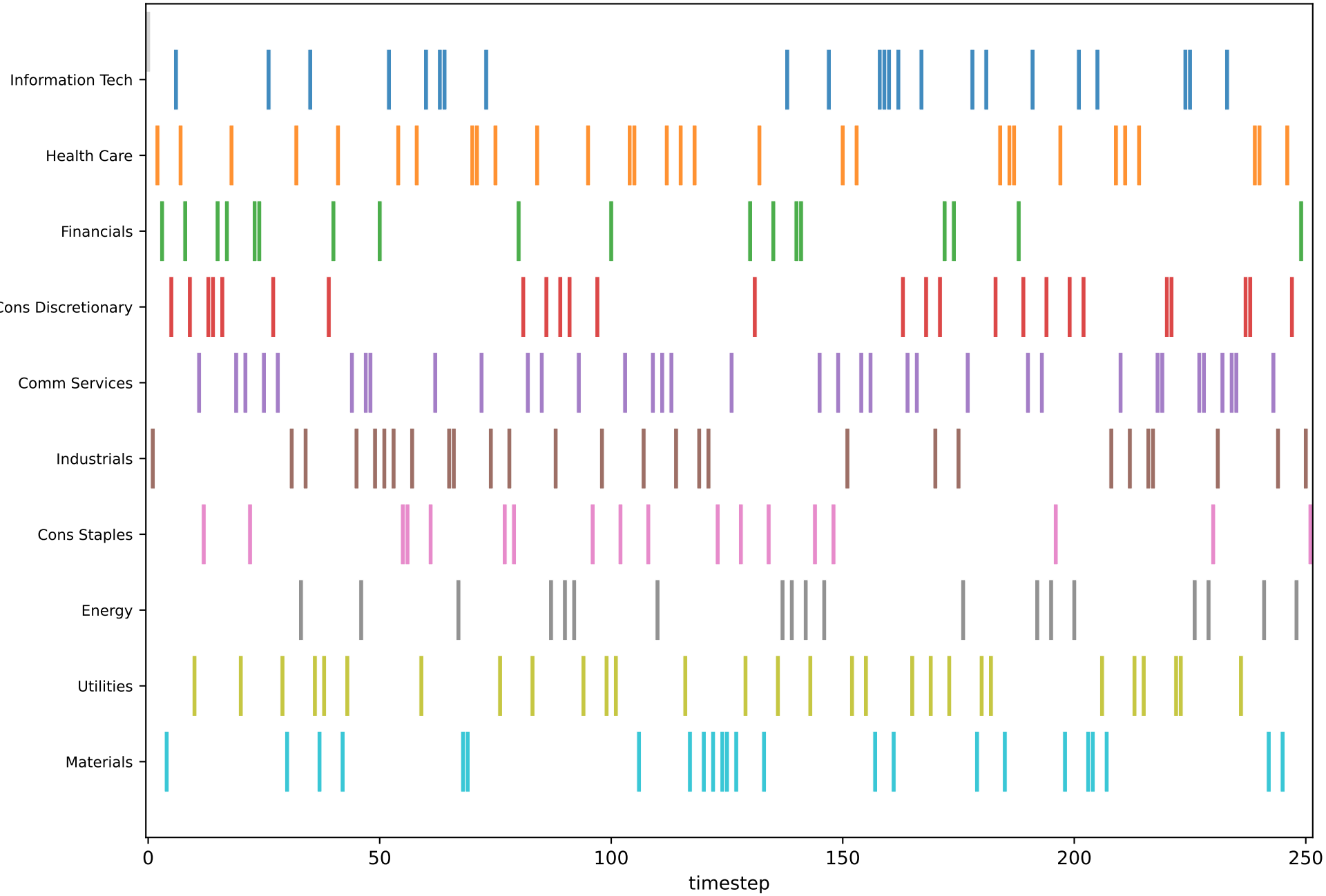


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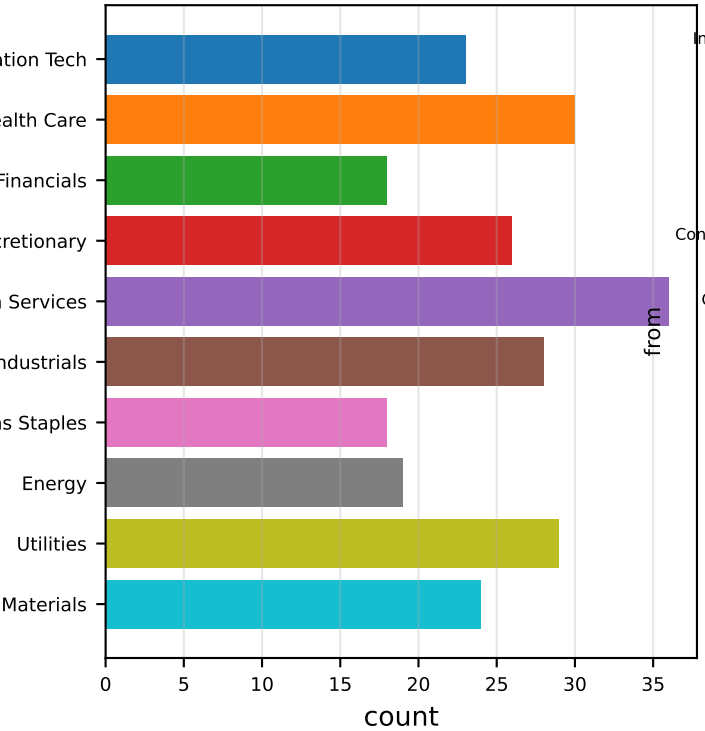
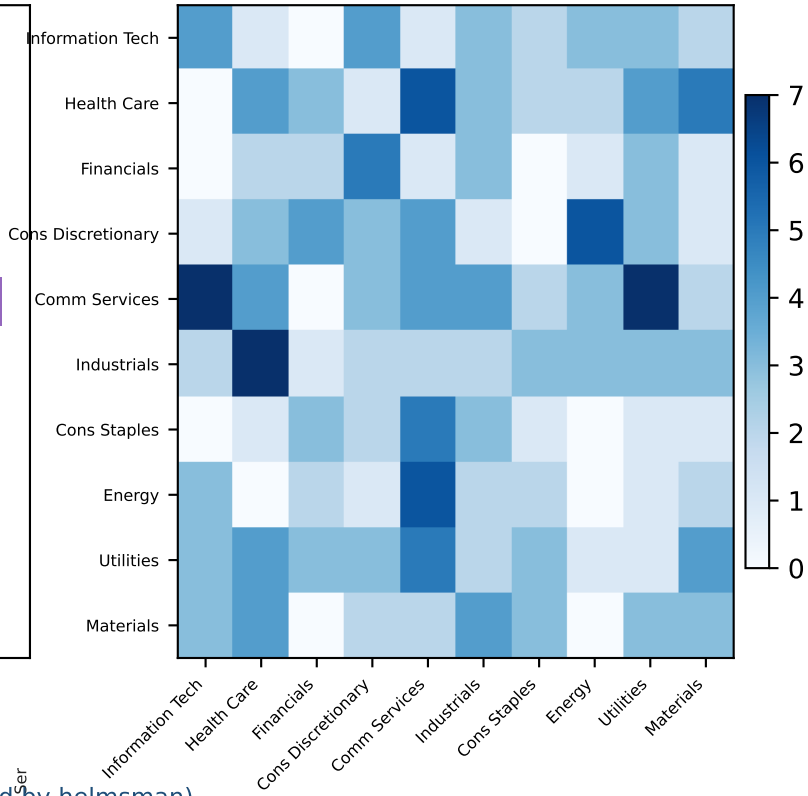
Plate 22 | Helmsman σ — frequency · transitions · per-year displacement

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

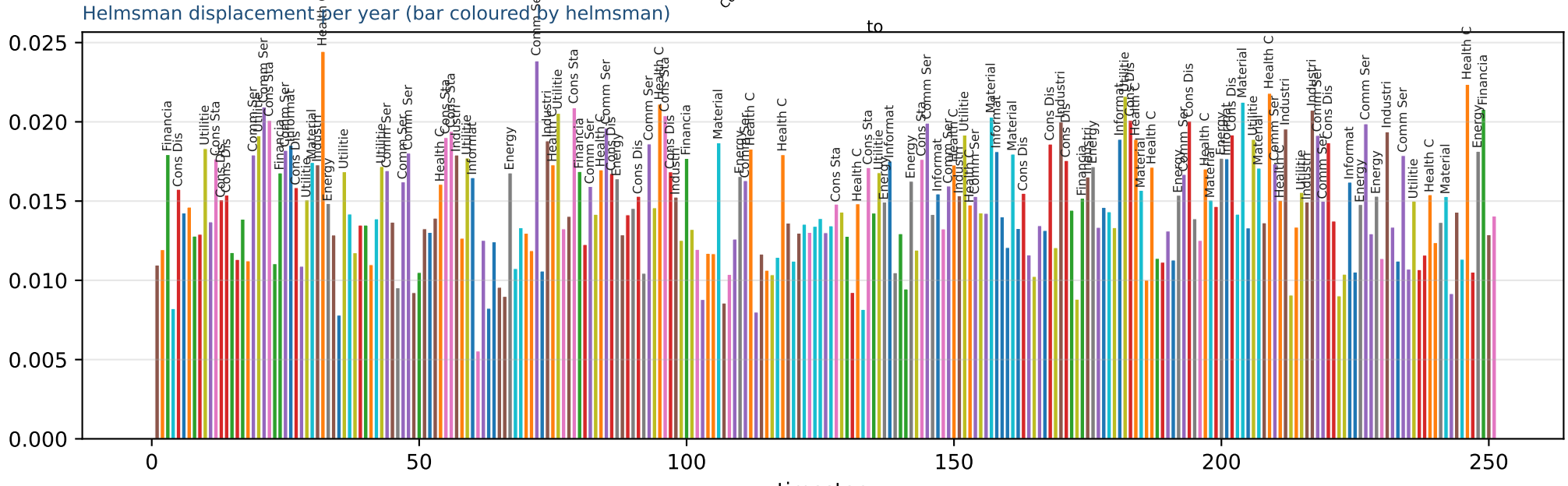
Helmsman frequency



Transition matrix



Helmsman displacement per year (bar coloured by helmsman)



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 23 | Ring class evolution + Hs

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

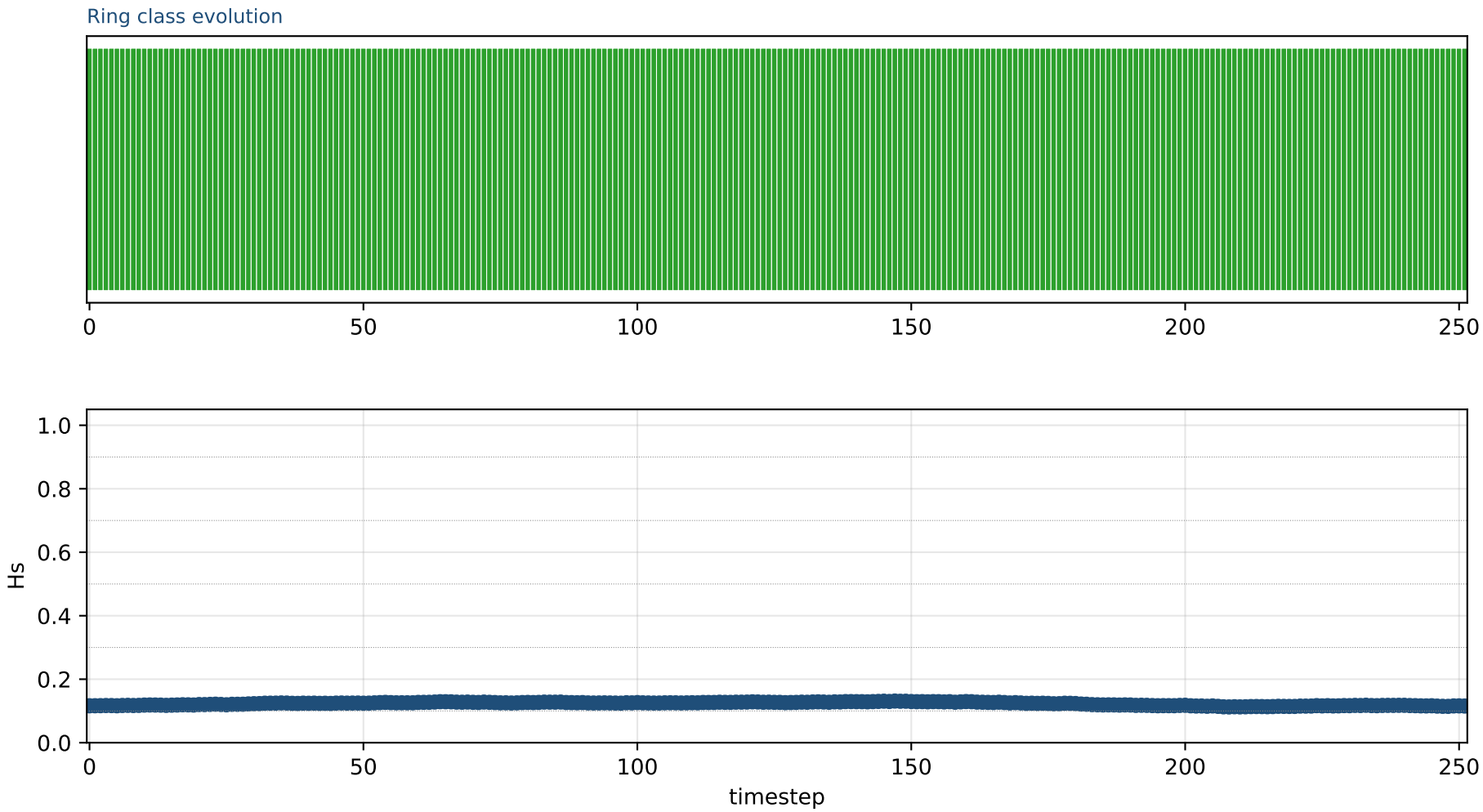


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Plate 24 | Lock event timeline (carriers × timesteps)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

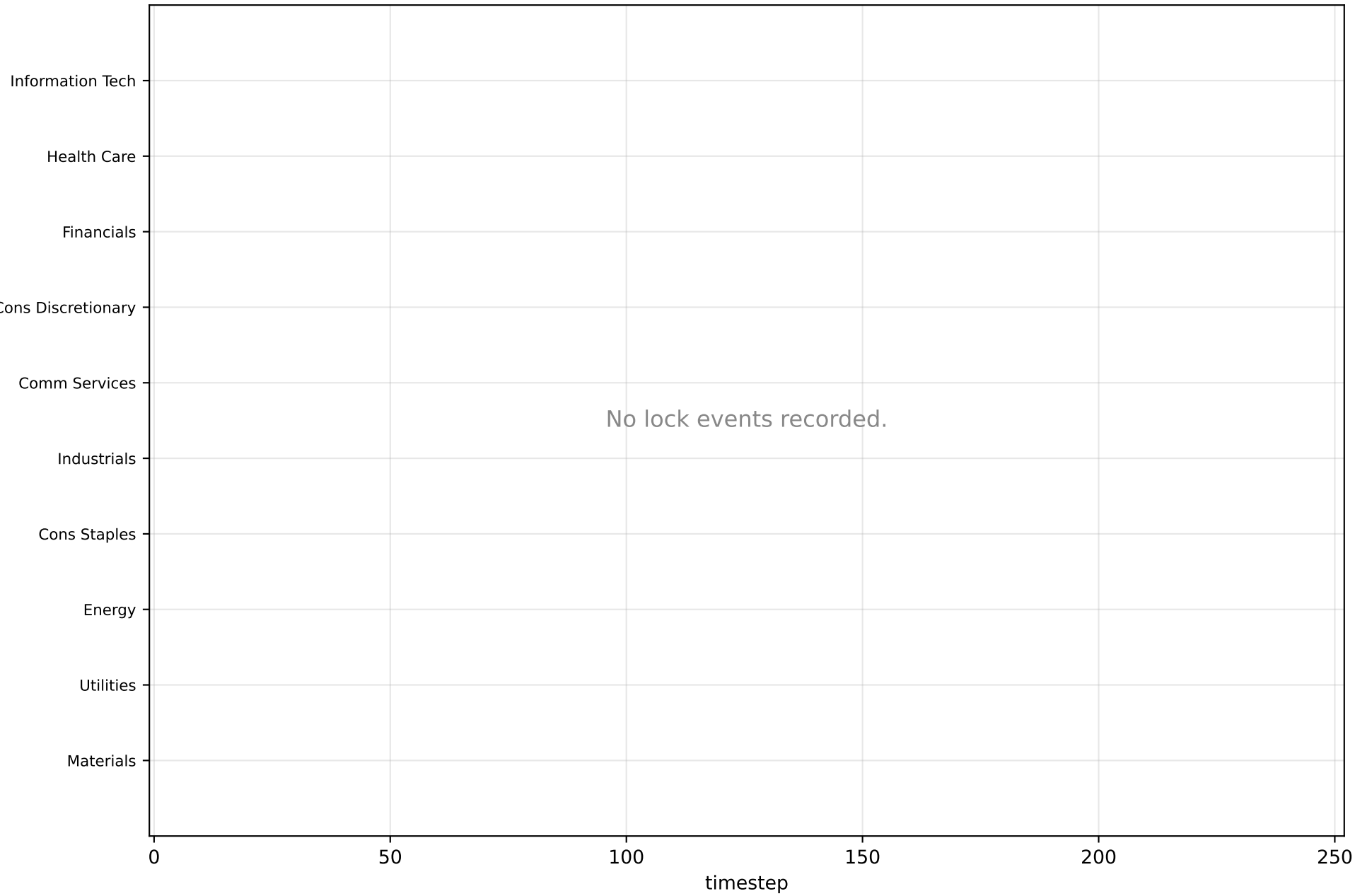

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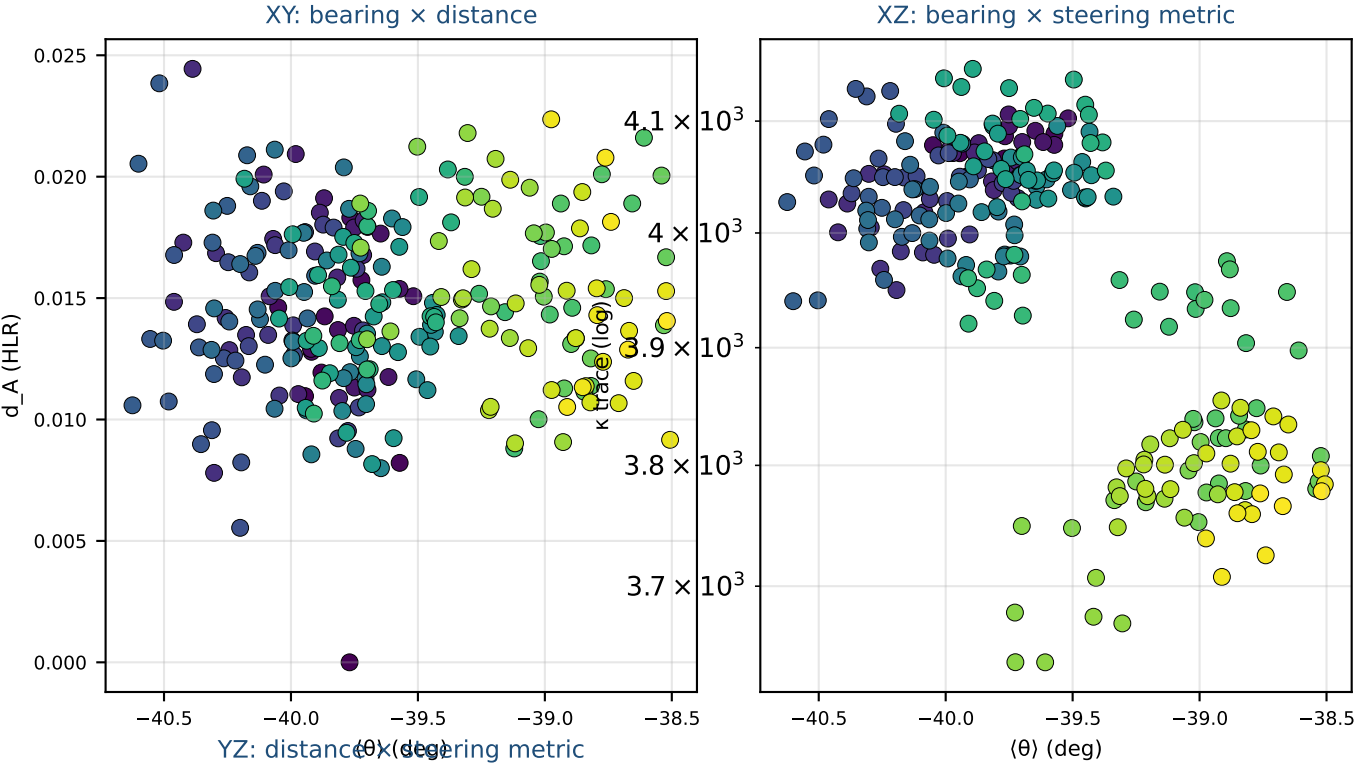
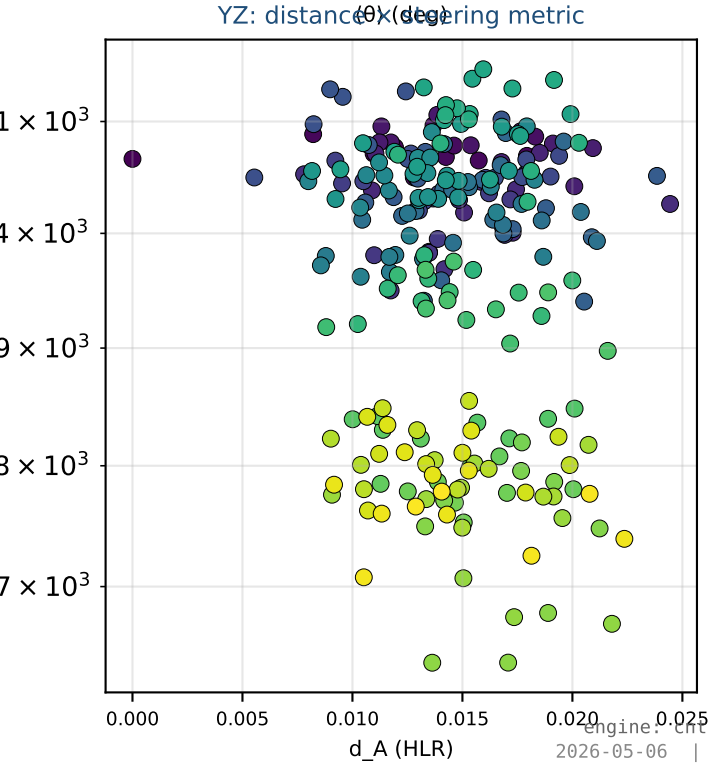
Plate 25 | Compositional Bearing Scope (CBS) — 3 orthogonal faces (Math Handbook Ch 16)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2





CBS info panel (Math Handbook Ch 16)

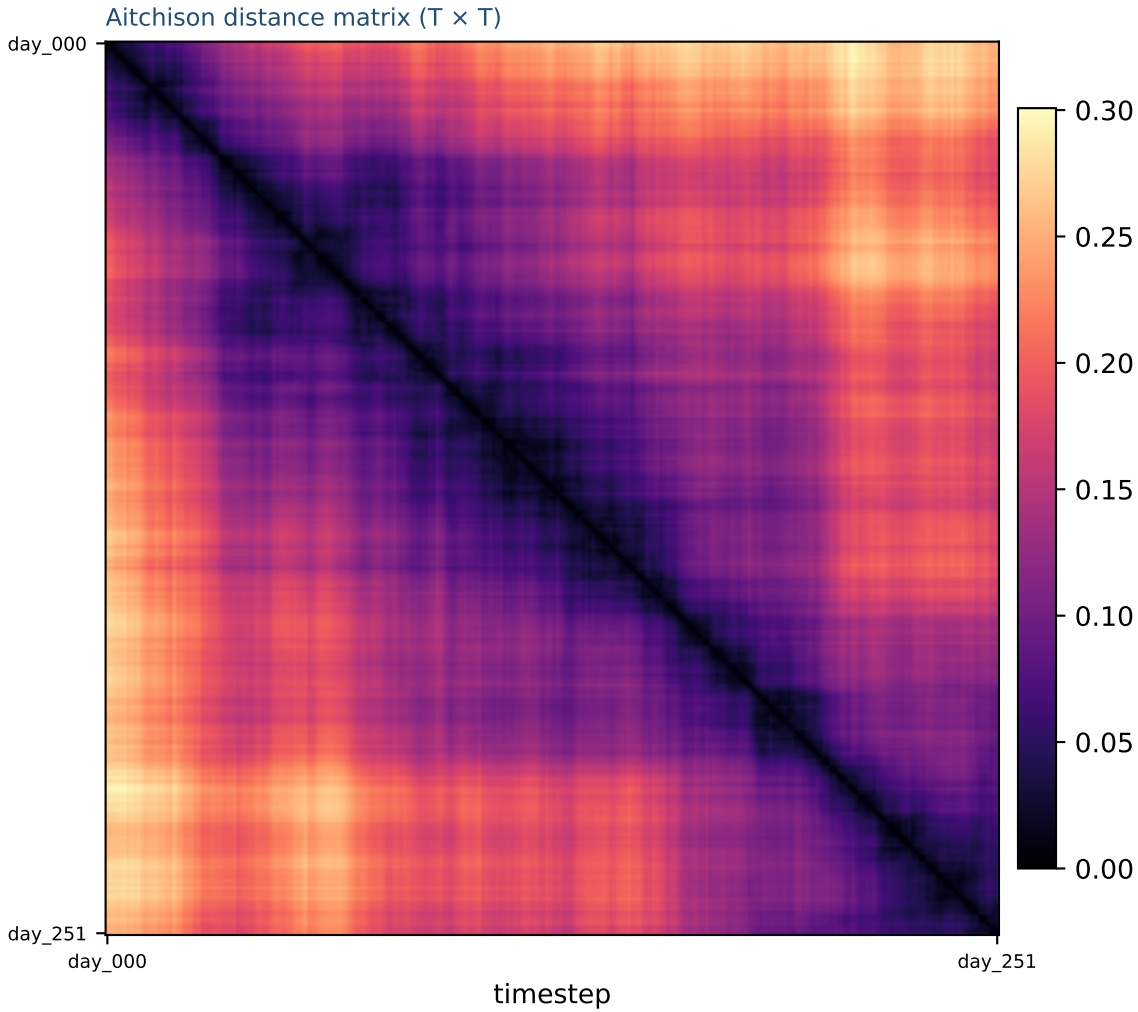
```
⟨θ⟩ range  : [-40.6, -38.5]°
d_A range  : [0.0000, 0.0244] HLR
κ range    : [3.64e+03, 4.15e+03]
Hs range   : [0.1127, 0.1312]
ω max      : 0.58°

Each face: one observation per timestep.
Colour = timestep (early=dark, late=bright).
Three faces locate the system in the
(bearing, distance, steering) measurement cube.
```

Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 26 | Pairwise divergence — distance matrix + top-15 ranking

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

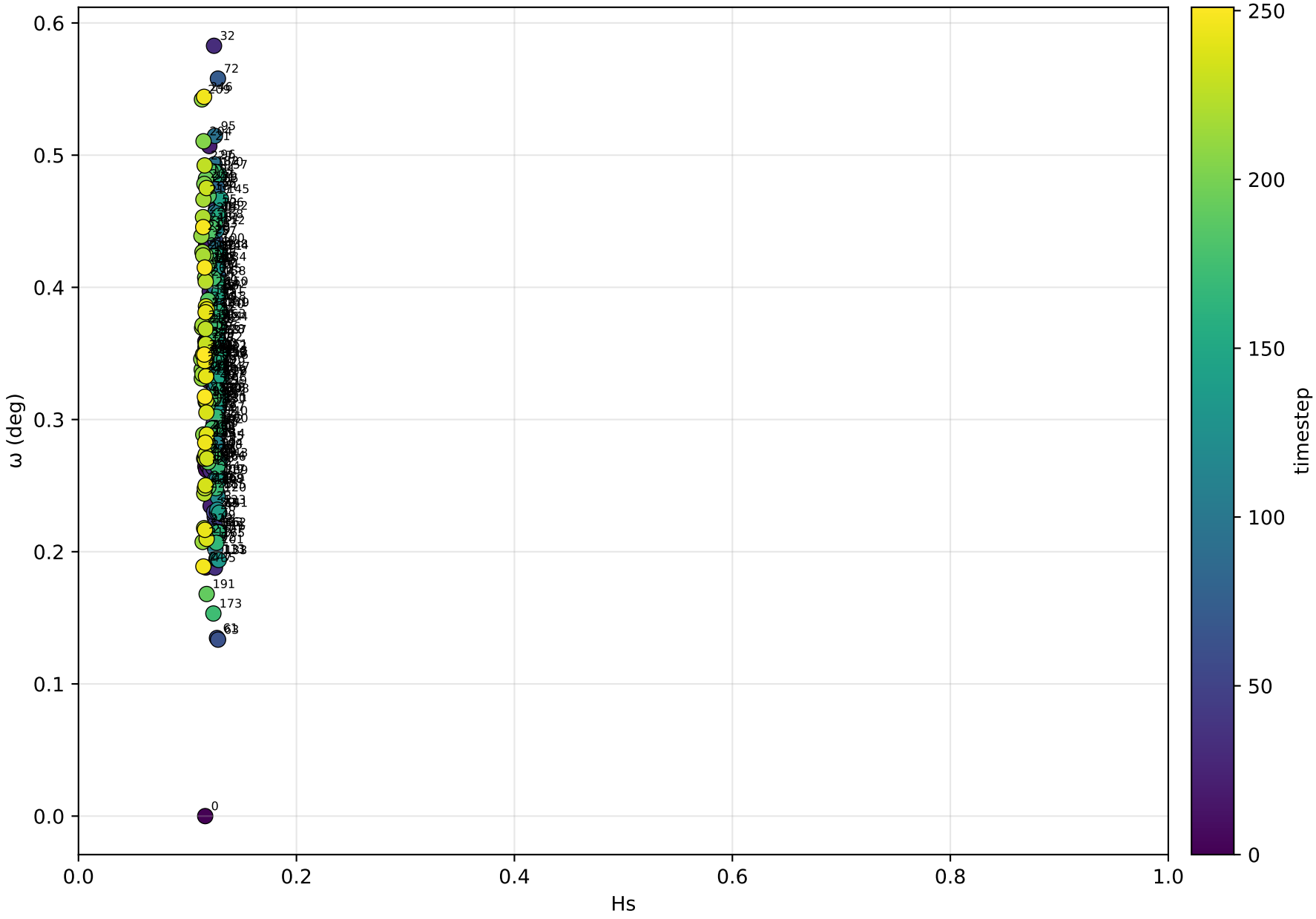


rank	A	B	d_A
1	day_005	day_210	0.3008
2	day_004	day_210	0.2988
3	day_001	day_210	0.2983
4	day_003	day_210	0.2981
5	day_002	day_210	0.2980
6	day_007	day_210	0.2952
7	day_000	day_210	0.2950
8	day_005	day_211	0.2949
9	day_009	day_210	0.2940
10	day_005	day_209	0.2938
11	day_004	day_211	0.2927
12	day_006	day_210	0.2924
13	day_003	day_211	0.2919
14	day_001	day_209	0.2919
15	day_004	day_209	0.2916

Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 27 | ω vs Hs scatter (XY test)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, S&P 500 GICS sector portfolio day_000 → day_251

Plate 28 | Stage 2 numeric summary

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

```
T = 252,  D = 10
carriers: Information Tech, Health Care, Financials, Cons Discretionary, Comm Services, Industrials, Cons Staples, Energy, Utilities

Navigation (System Course Plot):
  net_distance     = 0.2452 HLR
  path_length      = 3.6936 HLR
  course_directness= 0.0664

ω (deg)        : min 0.00  mean 0.34  max 0.58
Hs             : min 0.1127  mean 0.1228  max 0.1312
Aitchison norm : min 2.2574  mean 2.3356  max 2.3741
κ trace        : min 3638.34  max 4147.54
condition #    : min 1.24e+01  max 1.48e+01

Bearings: range [-142.01, 58.17]°,  std 49.27°
Lock events    : 0
Pairs          : 45
|r| > 0.7 pairs: 10

Helmsman frequency (top 5):
  Comm Services   36  (14.3%)
  Health Care     30  (11.9%)
  Utilities       29  (11.5%)
  Industrials     28  (11.1%)
  Cons Discretionary  26  (10.3%)

Variation matrix trace : 0.0000
```